



ENGIN 604 INTRODUCTION TO PYTHON FOR FINANCE — FALL 2021

HOMEWORK 1

Due: 11:59pm, Saturday, March 27

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The file `returns_portfolios.pkl` contains a nested list that stores the monthly return of two portfolios ($j = 1, 2$) from 2019-01-01 to 2020-12-01, plus an annualized risk-free rate in percentage terms with the same frequency. Using this file, answer the following:

1. Import the file `returns_portfolios.pkl` as shown below:

```
# imports the pandas library
import pandas as pd

# reads pickle
returns_portfolios = pd.read_pickle('returns_portfolios.pkl')
```

2. Compute the average return of each portfolio, defined as:

$$r_j = \frac{\sum_{i=1}^n x_i}{n}$$

3. Compute the standard deviation of each portfolio, defined as:

$$\sigma_j = \sqrt{\frac{\sum_{i=1}^n (x_i - r_j)^2}{n - 1}}$$

4. Compute the average of the risk-free rate. Since the risk-free rate is annualized, you must divide it by 12 to make it monthly and be consistent with the monthly return of the portfolios.
5. Compute the Sharpe ratio for each portfolio, defined as:

$$SR_j = \frac{r_j - r_f}{\sigma_j}$$

Where r_j is the average return of portfolio j , r_f the average of the risk-free rate not in percentage terms and σ_j the standard deviation of portfolio j .

6. Check whether the following statements are true:

- a. SR_1 is greater than SR_2 .
- b. SR_1 is less than SR_2 .
- c. The difference between SR_2 and SR_1 is positive.

7. Print to the terminal the Sharpe ratio for each portfolio using the following format:
`'The Sharpe ratio of portfolio j is {0:.4f}'`